Proof by Strong Induction

Soln 1 (b) Base case $n = 2t+1$

~~Induction hypothesis~~ Proof: for $t > 0$,
Binary Tree T and size of $T_i(n) \geq (2t+1)$
has vertex v such that subtree
rooted at v has size(h) $\geq t$
and $\leq 2t-1$

Proving base case: $n = 2t+1$

- Suppose vertex v has child a & b
  so, $a + b + 1 = 2t+1$
  $\Rightarrow a + b = 2t$
  for smallest $b = 1$, $a = 2t-1$
  and if $b > 1$, then $a < 2t-1$
  so $a \leq (2t-1)$

  if $b = t$ then $a = t$
  and that the largest value of b
  otherwise $a < b$ and $(2t+1)$
  node will be in subtree b.
  so since $a > b$, ~~then~~ and $b \leq t$
  so $a \geq t$.
Hence, vertex $a \geq t$ and $a \leq 2t-1$

Soln 1(b) Inductive hypothesis: Assume claim holds for every binary tree of ~~size k~~ of size k where $2t+1 \leq k < n$.

Now considering B Tree of size $n > (2t+1)$ ~~at height~~

~~At height~~ t

considering two subtree a & b where $a+b+1 = n$.

~~th~~ for $a > b$, $a \geq \frac{n-1}{2} \geq t$

so a can be,

1. $t \leq a \leq 2t-1$, so this the vertex (where $b = 1$ or $b(a)$)
2. $a \geq 2t+1$ so subtree below this vertex since $a < n$. due to hypothesis
3. $a \geq 2t$, in this case, going down subtree as well

~~3. a = 2t~~ ~~if b = t~~

Hence proved in every case T $n \geq 2t+1$ contains a vertex v whose rooted subtree has size ~~equal~~ ~~than~~. $t \leq$ ~~size~~ $\leq 2t-1$.

Que 2. Asymptotic Notation

Solution 2(a) i) $f(n) = 3\log_2^3 n$, $g(n) = n^2+1$

$(\log_2 n)^3$ grows much slower than n^2 since log itself grows much slower.

assuming $n = 1000$ using base 2

$\log_2(1000) \simeq 9.966$

so $f(1000) = 3(9.966)^3 \simeq 2969$

$g(1000) = 1000^2 + 1 = 1000001$

so ratio is $\frac{f(1000)}{g(1000)} \simeq 0.2971.$

if $n \to 1M$, the ratio is the ratio is almost "0"

so $\lim_{n\to\infty} \left(\frac{f(n)}{g(n)}\right) = 0$.

Hence, $f(n)$ is $o(g(n))$ (small o)

Soln 2 a) ii) for $n \geq 5$

$$f(n) = \underbrace{\binom{n}{0}}_{\text{choose 0}} + \underbrace{\binom{n}{1}}_{\text{choose 1}} + \underbrace{\binom{n}{2}}_{\text{choose 2}} + \cdots + \underbrace{\binom{n}{5}}_{\text{choose 5}}$$

$$f(n) = \sum_{k=0}^{\min(5,n)} \binom{n}{k}$$

~~1~~ $f(n)$ for $n \geq 5$

Since in subset order doesn't matter:

$$f(n) = 1 + n + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \frac{n(n-1)(n-2)(n-3)}{4!} + \frac{n(n-1)(n-2)(n-3)(n-4)}{5!}$$

so, $f(n) = 1 + n + \left(n^2 - \frac{n}{2}\right) + n^3 + n + (n^4 + n) + (n^5 + n)$

so $f(n) \geq c\, g(n^5)$

so $f(n)$ is $O(g)$ $(\exists g\ O)$

Asymptotic Notation

Soln a iii) $f(n) = 5^n$, $g(n) = n!$

n	5^n	$n!$	larger?
1	5	1	~~5~~ $f(n)$
5	3125	120	$f(n)$
25	2.98×10^{17}	1.55×10^{25}	$g(n)$
50	8.88×10^{34}	3.04×10^{64}	$g(n)$

so $\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0$

so $f(n)$ is $o(g(n))$ -small o

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Asymptotic Notation

Soln 2 a) iv) Prove / disprove

For all function $f, g: \mathbb{N} \to \mathbb{R}^+$

$f = O(g) \Rightarrow (g = o(f)$ or $f = o(g))$

This I can disprove if I chose

$f(n) = 2n+1$ and $g(n) = 2n+1$

so for $c = 1$, $f(n) = 1 \cdot g(n)$

that mean, $\frac{f(n)}{g(n)} = 1 \Rightarrow f \neq o(g)$

and $\frac{g(n)}{f(n)} = 1 \Rightarrow g \neq o(f)$

so I hence disprove.

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No. of late days previous pset: NA
No. of late days after including
this pset = 0.